



Administration of an herbal powder based on traditional Chinese veterinary medicine enhanced the fertility of Holstein dairy cows affected with retained placenta

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ABSTRACT

Herbal powder consisted of *Leonurus artemisia* (Laur.) S.Y. Hu F, *Angelica sinensis* (OLIV.) DIELS (radix), *Ligusticum chuanxiong* HORT (radix), *Sparganium stoloniferum* (Graebn.) Buch.-Ham.exJuz (radix), *Curcuma zedoaria* (Christm.) ROSC (radix), *Cyperus rotundus* Linn. (radix), and *Glycyrrhiza uralensis* FISCH (radix) has a high efficacy in facilitating earlier expulsion of the placenta and lowering puerperal metritis risk, and reproductive outcomes might represent the ultimate therapeutic goal of a treatment option for retained placenta in cows. This trial was carried out to further evaluate the effect of the herbal powder on the subsequent fertility of Holstein dairy cows with retained placenta. The animals were treated with herbal powder ($n = 85$, 0.5 g crude herb/kg bw, p.o.) once daily for 1–3 day(s) or ceftiofur hydrochloride ($n = 72$, 2.2 mg/kg bw, i.m.) twice daily for 3 consecutive days, and 75 cows with no clinically visible pathological conditions, given without assistance and with no retained placenta diagnosis were included into the control group. Calving-to-first-AI interval (67.2 ± 16.2 vs. 81.9 ± 28.7 , $p < 0.01$), services per conception (1.9 ± 1.2 vs. 2.4 ± 1.7 , $p < 0.05$), and calving-to-conception interval (101.8 ± 47.0 vs. 129.6 ± 56.6 , $p < 0.01$) were lower in the herbal group than in ceftiofur group. Percentage of cows pregnant within 180 days postpartum was the higher in cows from the herbal group compared to cows in ceftiofur group (75.0% vs. 90.6%, $p < 0.01$). Additionally, the cows in the herbal group had a very similar fertility compared to the healthy controls, including calving-to-first-AI interval (67.2 ± 16.2 vs. 66.3 ± 16.9 , $p > 0.05$), first AI conception proportion (48.2% vs. 50.6%, $p > 0.05$), services per conception (1.9 ± 1.2 vs. 2.0 ± 1.1 , $p > 0.05$), percentage of cows pregnant within 180 days postpartum (90.6% vs. 93.3%, $p > 0.05$) and calving-to-conception interval (101.8 ± 47.0 vs. 99.0 ± 44.7 , $p > 0.05$). Herbal powder used in this trial might have a beneficial clinical efficacy, and thus, might represent a potential effective treatment strategy to improve the subsequent fertility of dairy cows with retained placenta.

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1. Introduction

Reproductive efficiency is essential for the sustainability of the dairy enterprise and failures in reproduction would cause economic losses depend on drug cost, decreased pregnancy per artificial insemination, reduced conception, extended interval between calving and pregnancy and increased culling rate [1–3]. Previous

studies have shown that diseases related to the reproductive tract (retained placenta, metritis, and endometritis) are interrelated and can affect reproductive efficiency in dairy cows [4–6]. Particularly, retained placenta, an important postpartum uterine disease, is failure to pass the placenta within 24 h postpartum and cows with retained placenta have an increase risk of metritis or endometritis, and then decrease the subsequent fertility [4]. The accumulating evidence indicates that reproductive outcomes may be associated with better clinical parameters that could represent the ultimate therapeutic goal for retained placenta in dairy cows as well as in goats [4,7–12].

Many different treatment options, such as manual removal,

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intrauterine treatment or systemic administration of antibiotics, prostaglandin or estradiol, are applied to treat retained placenta, however, those treatments are described with a relative effectiveness, often with contradictory results or with a negative effect on cow's further fertility [11,13–15]. Traditional postpartum practices are believed to be beneficial in reducing the risk of postpartum diseases [16], and medicinal plants are often used to treat retained placenta in livestock [17–20]. Prophylactic strategy with use of *Sheng Hua Tang* immediately after delivery could lower retained placenta and puerperal metritis risk and improve overall reproductive efficiency of dairy herds [17,19]. Herbal remedies have beneficial effects for treat retained placenta, including facilitating earlier expulsion of the placenta and improving overall fertility in cows with retained placenta [8,20,21]. Our previous study [21] suggested that the herbal powder consisted of *Leonurus artemisia* (Laur.) S.Y. Hu F, *Angelica sinensis* (OLIV.) DIELS (radix), *Ligusticum chuanxiong* HORT (radix), *Sparganium stoloniferum* (Graebn.) Buch.-Ham.exJuz (radix), *Curcuma zedoaria* (Christm.) ROSC (radix), *Cyperus rotundus* Linn. (radix), and *Glycyrrhiza uralensis* FISCH (radix) has a high efficacy in hastening the separation and expulsion of the retained placenta and lowering the risk of occurrence of puerperal metritis in cows with retained placenta. Reproductive outcomes could represent the ultimate therapeutic goal of a treatment option for cows with retained placenta. Thus, we further evaluate the effect of the herbal powder for treatment retained placenta on the subsequent reproductive performance in Holstein dairy cows.

2. Material and methods

2.1. Herbal powder preparation

Herbal powder used in this trial was composed of 7 herbs—*Leonurus artemisia* (Laur.) S.Y. Hu F, 120.0 g, *Angelica sinensis* (OLIV.) DIELS (radix), 40.0 g, *Ligusticum chuanxiong* HORT (radix), 40.0 g, *Sparganium stoloniferum* (Graebn.) Buch.-Ham.exJuz (radix), 30.0 g, *Curcuma zedoaria* (Christm.) ROSC (radix), 25.0 g, *Cyperus rotundus* Linn. (radix), 25.0 g, and *Glycyrrhiza uralensis* FISCH (radix) 20.0 g—that was prepared as an herbal powder with 80–100 mesh particle size (150–180 μm). According to the procedure established by Cui et al. [21].

2.2. Animals

The present study was approved by the Institutional Animal Care and Use Committee of Lanzhou Institute of Husbandry and Pharmaceutical Sciences of the Chinese Academy of Agricultural Sciences (SCXK20008-0003). The trial was conducted in two large commercial dairy farms with similar management practices in Lanzhou city, China between August 2014 and December 2015. According to the experimental scheme of our previous studies [21], retained placenta was diagnosed as retention of fetal membranes at 12–24 h after delivery of the calf, and 157 cows with retained placenta were allocated to one of two groups based on their ear-tag numbers (the odd-numbered cows in the herbal group, and even-numbered cows in the ceftiofur group) and randomized block. And 75 cows with no clinically visible pathological conditions, given without assistance and with no retained placenta diagnosis were included into the control group in this trial. Particularly, the enrolled cows were 3–7 years old (2–5 lactations) and had a BCS (Body Condition Score) between 2.5 and 3.5 [22] during the peripartal stage. Cows were housed in free stall barns, received a total mixed ration [23] and had free access to high quality water. Cows were milked three times daily in computer controlled milking parlors and identified by ear tags and freeze marking.

2.3. Treatment protocol and clinical examination schedules

According to the treatment protocol described in our previous study [21], the animals received either an oral dose of 0.5 g herbal powder/kg bw once daily for 1–3 day(s) by drenching in the herbal group ($n = 85$) or systemic antibiotic treatment of 2.2 mg of ceftiofur/kg bw (Qilu Pharmaceutical co., LTD. Jinan, China) twice daily for 3 consecutive days in the ceftiofur group ($n = 72$). And the controls ($n = 75$) received no intervention. No cows affected with retained placenta were used in the untreated control group for ethical reasons.

In this trial, we focused on the effect of herbal powder treatment on the subsequent fertility of cows with retained placenta. After termination of voluntary waiting period ($VWP \geq 45$ days), all enrolled animals were monitored by the clinically visible oestrous signs and artificial insemination (AI) was performed by an experienced inseminator within 18–24 h after observed estrus. Pregnancy was confirmed by transrectal palpation and ultrasonography between days 36 and 42 following the last AI and confirmed by a second examination 30 days later. The cows were re-inseminated a maximum of 6 times (until day 305 post-partum) and no cows were culled for reproductive reasons before 305 DIM. Particularly, the breeding programs were similar and inseminations were performed throughout the year at the two farms in which the trial was conducted.

2.4. Data collection

To evaluate the reproductive performance of the cows treated with the herbal powder, the following data were collected from the farm records for each cow: lactation number, calving data, first AI, number of AIs and conception dates.

2.5. Efficacy measurements

The efficacy end point was the effect of herbal powder treatment on the subsequent fertility of cows with retained placenta, included calving-to-first-service interval, first AI conception proportion, service per conception, proportion of cows that were pregnant within 180 days postpartum and calving-to-conception interval.

2.6. Statistical analysis

All analyses were conducted in SPSS (version 17.0, IBM SPSS Statistics; New York, USA). The first AI conception proportion, proportion of cows that were pregnant at 305 DIM and service per conception were analyzed by logistic regression. The hazard of cows that were pregnancy during the first 305 days postpartum was analyzed by building Cox's proportional hazard model using forward likelihood-ratio testing. In the models, the time variable available was the interval in days from calving to pregnancy and the explanatory variable included twin calvings, dystocia, BCS, parity, retained placenta, puerperal metritis and milk fever. All cows that were not pregnant or never inseminated at 305 DIM were censored. Additionally, specific censored cases were cows that became pregnant, but then aborted and did not become pregnant again within 305 DIM. Interactions between explanatory variables and time, and survival curves were used to evaluate the proportionality of hazard rate. The level of significance was set at $p = 0.05$.

3. Results

The herbal powder affected on reproductive performance of cows with retained placenta (Table 1). Calving-to-first-AI interval

Table 1

Influence of herbal treatment on the postpartal fertility in cows with retained placenta.

Parameters	Herbal group	Ceftiofur group	Control group
Calving-to-first-AI interval, days	67.2 ± 16.2 ^B	81.9 ± 28.7 ^A	66.3 ± 16.9 ^B
First AI conception proportion, (%)	48.2	34.7 ^a	50.6 ^b
Service per conception	1.9 ± 1.2 ^b	2.4 ± 1.7 ^a	2.0 ± 1.1 ^b
Proportion of cows that were pregnant within 180 days postpartum, %	90.6 ^B	75.0 ^A	93.3 ^B
Calving-to-conception interval, days	101.8 ± 47.0 ^B	129.6 ± 56.6 ^A	99.0 ± 44.7 ^B

Values within the same row marked with different letters in superscript differ significantly: a, b = $p < 0.05$; A, B = $p < 0.01$.

and calving-to-conception interval were lower in herbal group than in ceftiofur group ($p < 0.01$). Calving-to-first-AI interval were different between the cows in herbal and ceftiofur group, as presented in Fig. 1 ($p = 0.009$). Herbal treatment effectively reduced the number of services per conception compared to ceftiofur treatment ($p < 0.05$). Percentage of cows pregnant within 180 days postpartum was the higher in cows from herbal group compared to cows in ceftiofur group ($p < 0.01$). Survival curves were different between the cows in herbal group and ceftiofur group in terms of the interval-to-conception ($p < 0.01$; Fig. 2). In addition, the cows treated with herbal powder had a very similar reproductive performance compared to the healthy controls.

4. Discussion

According to the working hypothesis of the present study, herbal powder treatment could enhance the reproductive performance of cows with retained placenta. More recently, we have demonstrated that the herbal powder could improve the uterine environment in dairy cows affected with retained placenta through an earlier placental detachment and a lower risk of occurrence of puerperal metritis [21]. Thatcher et al. [24,25] have suggested that a more optimal restoration of the uterine environment could advance the occurrences of early and frequent estrus after parturition, and thus increase fertility in cows. In this study, the herbal powder treatment advanced days to first AI, reduced the service per

conception and shorten days open in cows with retained placenta. The findings tend to suggest that more cows resumed cyclicity and exhibited good uterine recovery sooner after calving. Additionally, more cows treated with the herbal powder become pregnant within 180 days postpartum, and exhibited a very similar reproductive performance compared to the healthy controls. The pregnancy rate is a good overall measure of reproductive performance. The results demonstrated that suggested that the herbal powder treatment improved the overall fertility of cows with retained placenta. This is in accordance with studies that compared similar treatment strategies [8,20]. Therefore, the herbal powder treatment might have great potential for the medical management of retained placenta in dairy cows.

Generally, the consequences of retained placenta on the fertility of a herd are partly dependent on when puerperal metritis occurs [26–28] and intrauterine or systemic treatment with ceftiofur hydrochloride have often been applied to control uterine infection secondary to retained placenta in dairy cows due to no antibiotic residues in the milk when used according to the label directions [7,11,12,29]. Our previous study showed that ceftiofur treatment for retained placenta could reduce the puerperal metritis risk [21], however, did not improve the fertility of cows compared to the herbal powder in this trial. The results might be related to the synergistic effect of the herbs in this herbal powder based on traditional Chinese veterinary medicine. In the herbal powder, a large dose of *Leonurus artemisia* (Laur.) S.Y. Hu F was believed to

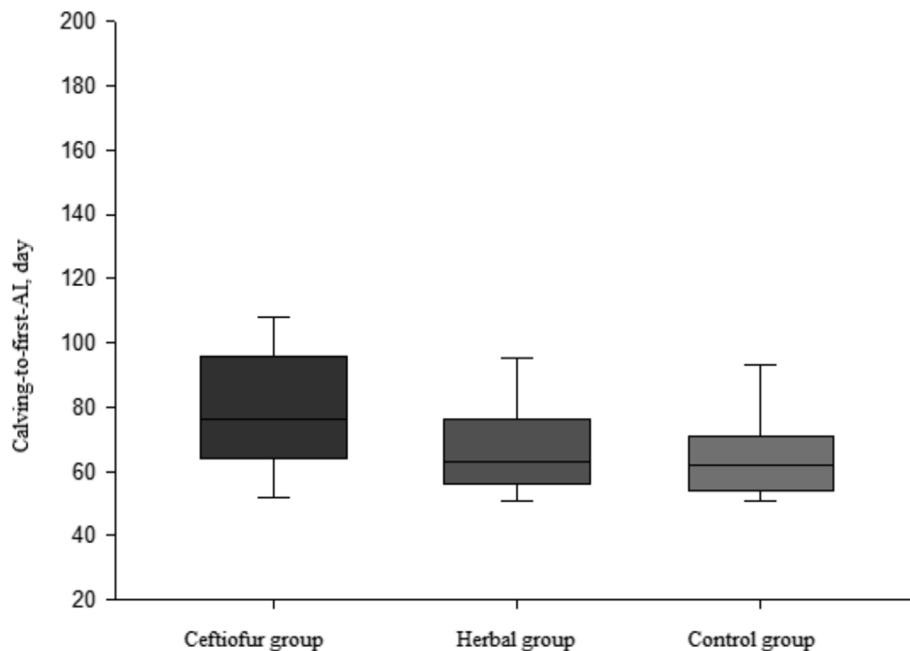


Fig. 1. Interval to first AI of box plots indicate the median time from calving to first AI with 25th and 75th percentile values at the bottom and top of each box in ceftiofur, herbal and control groups (outliers and extremes not shown).

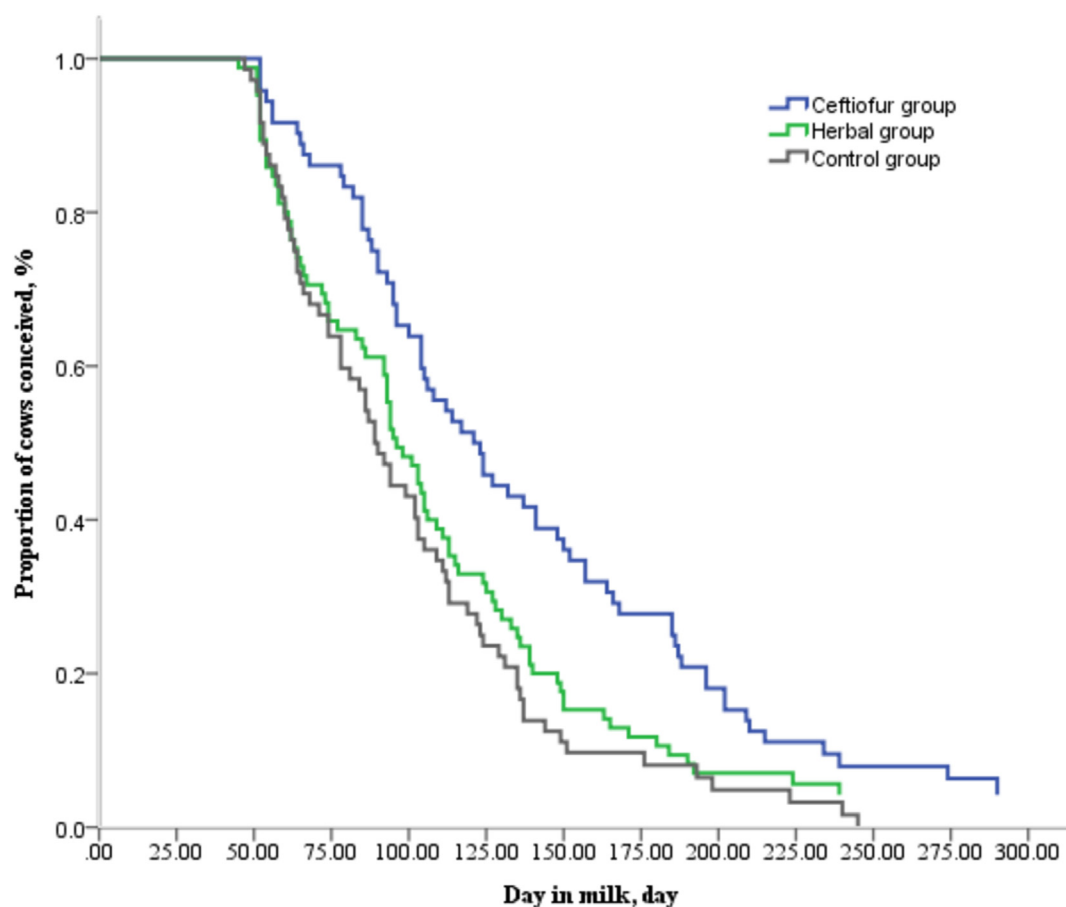


Fig. 2. Survival curves for interval to pregnancy in cows affected with retained placenta treating with an oral dose of 0.5 g herbal powder/kg bw once daily for 1–3 day(s), ceftiofur hydrochloride (2.2 mg/kg bw i.m.) twice daily for 3 consecutive days, or healthy cows receiving no any intervention.

improve uterus conditions by modulate uterine hemorheology and microcirculation [30]. *Angelica sinensis* (OLIV.) DIELS (radix)-*Ligusticum chuansiong* HORT (radix) can produce synergistic and complementary effects upon hematopoietic functions and promoting blood stasis [31–35]. *Glycyrrhiza uralensis* FISCH (radix), a tonifying herbal medicine extensively used in traditional Chinese veterinary medicine, modulates host immunity through anti-inflammatory, immunomodulatory and anti-oxidative properties [36]. Based on the above potential pharmacological activities of single herb, the herbal powder seemed not only to improve the uterine condition but also improved the overall physical health of cows with retained placenta. Additionally, herbal powder treatment could improve the uterine environment duo to an earlier placental detachment and a lower puerperal metritis risk. And thus the herbal powder is beneficial for improving the subsequent fertility of cows with retained placenta. Public awareness of the prudent use of antibiotics in the dairy and food industry is steadily improving and the demand for alternative therapies is increasing. From a clinical perspective, the herbal powder might have high potential for the medical management of dairy cows with retained placenta to improve their subsequent fertility.

5. Conclusions

The present results provided evidence for a beneficial clinical effect in improving the subsequent fertility of cows with retained placenta, including shorter calving-to-first-service interval and calving-to-conception interval, less service per conception and

higher proportion of cows that were pregnant within 180 days postpartum. Thus, this herbal remedy might represent a potential effective treatment strategy to improve the subsequent fertility of cows with retained placenta. Future investigations will focus on the broader involvement of the herbal powder in health-related quality of life for cows with retained placenta and the mechanisms underlying its therapeutic effects.

Conflicts of interest

None of the authors has any financial or personal relationships that could inappropriately influence or bias the content of the paper.

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